SuperDARN Data & Publication Sources

Documentation of metadata resources, search indices, coordinate models, and filtering criteria.

1. Radar Location & Hardware Coordinates

The coordinates (geographic coordinates and operating altitudes) of the SuperDARN radar stations are retrieved from the official pyDARN library database, which references the authoritative community hardware registry:

- **Source Registry:** SuperDARN Hardware Database (SuperDARN/hdw)
- **Retrieval Method:** Programmatically parsed using python's pydarn.utils hardware modules, extracting the earliest coordinates for each station.
- **Hemisphere Classification:** Categorized based on geographic latitude (Northern hemisphere for positive values, Southern hemisphere for negative values).

2. Altitude-Adjusted Corrected Geomagnetic (AACGM) Model

Because the Earth's magnetic field changes over time and has non-dipolar components, geographic coordinates do not line up with magnetic coordinates. We converted the geographic coordinates using the following models:

- **Coordinate Engine:** AACGM-v2 (Altitude-Adjusted Corrected Geomagnetic coordinates, Version 2) C-library implementation.
- **Python Library:** aacgmv2 (Version 2.7.1), wrapping the standard geomagnetic coordinate transforms.
- **Epoch Settings:** The AACGM coordinates are computed dynamically for the specific year each radar went operational to capture the secular variation of the magnetic field.

3. Scientific Publication & Citation Metrics

Publication counts were programmatically parsed from the official SuperDARN Canada registry. Corresponding citation metrics were retrieved directly from the Crossref scholarly metadata index using the DOIs registered by the network:

- **Publication Database:** SuperDARN Canada Official Publications Registry (https://superdarn.ca/publications)
- **Citation Database:** Crossref Metadata Search API (https://api.crossref.org/works)
- **Retrieval Method:** Programmatically parsed the registry's CSV, extracted DOIs, and batch queried Crossref in groups of 50 using filter queries to fetch is-referenced-by-count metadata.

- **Citation Cohorts:** Citations are associated with the publication year of the papers. The cumulative curve shows the total citations of papers published up to each year.

4. Summary of Data Sources

Dataset / Feature	Primary Source	Provider / Institution	Access Link / API
Radar Station Metadata	SuperDARN hdw Registry	SuperDARN Data Simulator Working Group	github.com/SuperDARN/hdw
Geomagnetic Coordinates	AACGM-v2 Model	SuperDARN Collaboration / NASA	via python aacgmv2 wrapper
Publications Registry	SuperDARN Canada Portal	University of Saskatchewan	superdarn.ca/publications
Citation Metadata	Crossref Index	Publishers International Linking Association	api.crossref.org/works

5. Visualization Variants

Two separate publication and citation plots have been compiled:

- **Hybrid Registry & Crossref Plot:** Uses the official 1,303 peer-reviewed paper registry from SuperDARN Canada as the source of publications (bars), and queries the Crossref API using their registered DOIs to compile the cumulative citation totals.
- **Crossref-Only Plot:** Restricts both publication counts and citation metrics solely to the Crossref database. The query searches for works matching the term 'SuperDARN' (yielding 414 total publications and 6,926 cumulative citations by 2025).